

Exp 7: Configuration Management with Ansible: Basics of Ansible: Inventory, Playbooks, and Modules, Automating Server Configurations with Playbooks, Hands-On: Writing and Running a Basic Playbook.

Step 1: Introduction to Ansible

What is Ansible?

Ansible is an open-source IT automation and configuration management tool. It allows you to manage multiple servers and perform tasks such as:

- Configuration Management: Automate the configuration of servers.
- Application Deployment: Deploy applications consistently.
- Orchestration: Coordinate complex IT workflows and processes.

Key Concepts in Ansible

- Inventory: An inventory is a file (usually in INI or YAML format) that lists the hosts (or groups of hosts) you want to manage. It tells Ansible which machines to target.
- Playbook: A playbook is a YAML file that defines a set of tasks to be executed on your target hosts. It is the heart of Ansible automation. In a playbook, you specify:
 - Hosts: The target machines (or groups) on which the tasks should run.
 - Tasks: A list of actions (using modules) that should be executed.
 - Modules: Reusable, standalone scripts that perform specific actions (e.g. installing packages, copying files, configuring services).
- Modules: Ansible comes with a large collection of built-in modules (such as apt, yum, copy, service, etc.). These modules perform specific tasks on target hosts. You can also write custom modules if needed.

Why Use Ansible?

- Agentless: Ansible uses SSH to communicate with target hosts, so no agent needs to be installed on them.
- Simplicity: Playbooks use simple YAML syntax, making them easy to write and understand.
- Idempotence: Ansible tasks are idempotent, meaning running the same playbook multiple times yields the same result, ensuring consistency.
- Scalability: Ansible can manage a small number of servers to large infrastructures with hundreds or thousands of nodes.

Step 2: Install Ansible on Ubuntu

- `sudo apt update`
- `sudo apt install ansible`
- `ansible --version`

Step 3: Ansible Inventory

1) Create an inventory file

`nano inventory.ini`

2) Add the following content

`[local]`

`localhost ansible_connection=local`

Run on terminal:

- `ansible all -m ping -i inventory.ini`
- `ansible localhost -m apt -a "name=nginx state=present" --become`
- `ansible localhost -m command -a "uptime"`

Step 4: Write playbook

`nano setup.yml`

Add the following:

- name: Configure Local Machine

hosts: local

become: yes

tasks:

- name: Update package lists

apt:

update_cache: yes

- name: Install Nginx

apt:

name: nginx

state: present

- name: Start Nginx service

service:

name: nginx

state: started

Step 5: Run playbook:

`ansible-playbook -i inventory.ini setup.yml`

run on terminal:

`systemctl status nginx`

Step 6: Open web browser and type:

`http://localhost`

Output:

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.